UNIVERSITY OF PERADENIYA
DEPARTMENT OF PHYSICS

END SEMESTER EXAMINATION - SEMESTER II - 2016/2017
PH 102 - General Physics II

Time allowed: ***Three hours***

Answer **Five** questions selecting **Two** from Section **A**, **Two** from Section **B** and **One** from Section **C**.

(Use of calculators are allowed)

Electron charge (e)	$= 1.60 \times 10^{-19}$ C
Rest mass of electron (m_o)	$= 9.11 \times 10^{-31}$ kg
Permittivity of free space (ε_o)	$= 8.85 \times 10^{-12}$ F m^{-1}
Permeability of free space (μ_o)	$= 4\pi \times 10^{-7}$ H m^{-1}
Planck constant (h)	$= 6.63 \times 10^{-34}$ J s
Speed of light in vacuum (c)	$= 3.00 \times 10^{8}$ m s^{-1}
Boltzmann constant (k)	$= 1.38 \times 10^{-23}$ J K^{-1}
Avogadro's number (N_A)	$= 6.02 \times 10^{23}$ mol^{-1}.

SECTION - A

1. (a) Write down the Gauss' law in words and give it as an expression.

   (b) Give the relationship between the electric field intensity $E(r)$ and the potential $V(r)$.

   (c) Suppose that charge Q is uniformly distributed in a solid sphere of radius R.

   (i) Obtain an expression for the electric field intensity at distance r from the centre of the sphere for $r \leq R$ and $r > R$.

   (ii) Sketch the graphs of $E(r)$ versus r and, $V(r)$ versus r for the charge distribution from $r = 0$ to infinity.

   (d) Using the expression in part (c)(i), find the potential difference between the centre and the surface of the sphere.

   (e) The atomic nucleus can be considered as a spherical shape having radius $R = 10^{-15}$ m and uniformly distributed charge of $Q = 1.6 \times 10^{-19}$ C. Find the potential difference between the centre and the surface of the nucleus.

2. (a) A particle of charge q and mass m moves at constant speed v perpendicular to a uniform magnetic field of flux density B. Using the given parameters determine the path of the charge particle.

   (b) Using a clear labeled diagram, explain the action of a cyclotron.

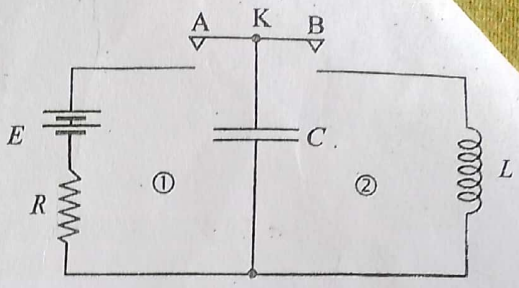
   (i) Obtain an expression for the cyclotron frequency.

   (ii) The maximum radius of a cyclotron is 0.5 m and the magnetic flux density is 0.25 T. What should be the operating frequency of the cyclotron to obtain a beam of α-particles? The mass of an α-particle is 6.65×10^{-27} kg.

   (iii) Calculate the maximum kinetic energy of α-particles (in MeV) that can be obtained from the cyclotron in part (ii).

   (iv) The maximum kinetic energy of any cyclotrons is limited due to two practical reasons. Briefly explain them.

3. As shown in the figure, a capacitor of capacitance C, a resistor of resistance R, an inductor of inductance L, two way key K and a cell of e.m.f. E with negligible internal resistance are connected to make two circuits ① and ②.



(a) Initially end A of the key K is pressed at time $t = 0$ to charge the capacitor by closing circuits ①.

(i) Obtain an expression for charge q in the capacitor after time t.

(ii) Sketch a graph of q versus t for the capacitor.

(iii) Define the time constant of the circuit ①.

(iv) If $E = 6.0$ V and $C = 8$ mF, find the value of the resistance R to obtain a time constant of 0.5 s^{-1} for the circuit ①.

(b) Once the capacitor in the circuit is fully charged to its maximum theoretical capacitance, the key K is pressed to end B to close circuits ② and starts to measure time from $t = 0$.

(i) Obtain an expression for the current i in the circuit ② at any time t.

(ii) Sketch the graph of i versus t for the circuit ②.

(iii) If $L = 2\times10^{-3}$ H, find the maximum curent and resonant frequency of the circuit ②.

SECTION B

4. (a) State one application of Bohr theory.

(b) Write down an equation showing the Bohr's postulate related to the angular momentum of the electron. Name any symbols used.

(c) Using the equation written in part (b) and the de Broglie's equation, show that the perimeter of a Bohr orbit of radius r is an integer multiple of the electron wavelength.

(d) Assuming the nucleus mass is very large, write down an expression for the total energy of a H-atom. Identify the terms in the total energy expression and name the symbols used.

(e) The quantized electron energy levels of a H-atom are given by

$$E_n = -\frac{me^4}{8\epsilon_0{}^2h^2}\frac{1}{n^2} = \frac{-13.6}{n^2} \text{ (in eV)}$$

where n =1, 2, 3, 4,..... and other symbols have their usual meanings. With the help of the above expression

(i) calculate the ionization energy (in eV) of doubly ionized Li and

(ii) calculate the shortest and the longest wavelengths of the Balmer series of doubly ionized Li.

(f) Give two failures of Bohr theory.

5. (a) (i) Briefly explain the photoelectric effect and state the important concept explained using the above phenomena.

(ii) A surface of zinc is illuminated by a beam of photons and photoelectrons are observed. If the work function of zinc is 4.31 eV, what is the largest wavelength that can emit photoelectrons?

(iii) What is the stopping potential when light of wavelength 220 nm is used?

(b) (i) Briefly explain the Compton shift.

(ii) Compton shift is given by the equation $\lambda' - \lambda = \frac{h}{mc}(1 - cos\theta)$,

where the symbols have their usual meanings. The incident X-ray and γ-ray wavelengths are λ_X and λ_γ. Qualitatively sketch the graphs of λ' versus (1-cosθ) for X-rays and γ-rays in the same diagram.

(c) The Young's double slit experiment can be performed using a beam of electrons. What is the importance of this experiment?

When the two slits are separated by a distance of 2.0×10^{-6} m, the first order bright fringes are observed on the screen at an angel 1.6×10^{-4} rad from the midpoint of the two slits.

(i) Calculate the de Broglie's wavelength of the electron beam.

(ii) Calculate the kinetic energy of an electron in the beam.

(iii) If a slit has a width of 1.5×10^{-11} m, estimate the minimum uncertainty of the speed of the electron.

6. (a) (i) Write down the law of radioactive decay.

(ii) Suppose that there are N_0 number of radioactive nuclei in a sample. Obtain an expression for the number of radioactive nuclei, $N(t)$, in this sample after time t.

(iii) Define half-life and obtain the relationship between the half-life and the decay constant.

(iv) The ratio of ^{14}C to ^{12}C in a sample of living wood is $1:10^{12}$. A piece of wood from an archeological site gave a total count rate of 20 disintegrations per minutes due to decay of ^{14}C. Estimate the age of the piece of wood. The half-life of ^{14}C is 5730 years.

(b) (i) Give the purpose of the moderator, the control rods and the coolant used in a nuclear reactor.

(ii) Find the value of x and y for the following nuclear reaction.

$$^{235}_{92}U + n \rightarrow {}^{95}_{42}Mo + {}^{139}_{57}La + xn + y\beta^-.$$

(iii) Neglecting the mass of β^- particles, calculate the total energy released by 1.0 g of $^{235}_{92}U$ in the above nuclear reaction.

$^{95}_{42}Mo = 94.906$ u

$^{139}_{57}La = 138.906$ u

$^{235}_{92}U = 235.044$ u

n = 1.009 u

1 u = 931 MeV/c^2

(iv) Suppose that the mean energy released per $^{235}_{92}U$ atom in a typical nuclear reactor is same as the value calculated above. If the reactor consumes 0.5 kg of pure $^{235}_{92}U$ in one year, and the efficiency of generation of electricity is 40%, find the mean electrical power output of the reactor.

SECTION – C

7. (a) Give two **solid materials** as examples for each of the following bonding types.
(i) Ionic (ii) Covalent (iii) Metallic

(b) Give two examples for each of the following intermolecular bonding types.
(i) Van der Waals (ii) Hydrogen

(c) Qualitatively sketch the potential energy curves in the same diagram for the following materials. Name the axes.

Material	Melting point (°C)	Thermal expansion ($\times 10^{-6}$ °C^{-1})	Force constant ($\times 10^{9}$ N m^{-1})
Polyethylene	65	100	0.8
Copper	1085	20	117

(d) The interaction potential between a pair of neutral atoms as a function of distance (r) is given by,

$$U(r) = \left[\frac{\sigma}{r}\right]^3 - \frac{\sigma}{r}$$, where σ is a constant.

(i) Obtain an expression for the force between the above atoms as a function of r.
(ii) Find the equilibrium bond length, r_0, of the two atoms in terms of σ ~~and n~~.
(iii) Determine the binding energy of the system in terms of σ ~~and n~~.

8. (a) (i) The nuclei of the O_2 molecule are separated by a distance 1.20×10^{-10} m. The relative atomic mass of oxygen is 16. If the molecule is in its first excited state of the rotational energies, calculate its rotational (angular) frequency.

(ii) Calculate frequency of the emitted photon when the molecule transits from the first excited state to its ground state.

(b) (i) The vibrational frequencies of CO and NO are 6.42×10^{13} Hz and 5.63×10^{13} Hz, respectively. Which molecule has the weaker bond? Show the calculations related to the answer.

(ii) Compare the vibrational energy levels of heavy hydrogen (D_2) molecules with those of H_2 molecules and comment on the differences.

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